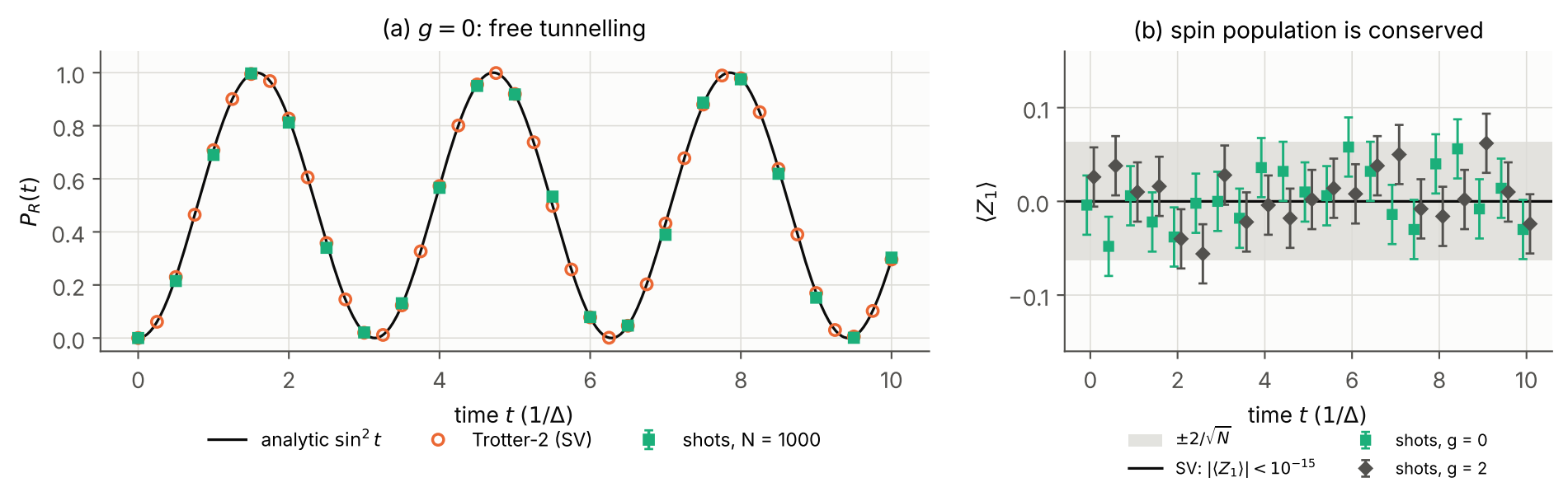
(a) $g = 0$: free tunnelling
$P_R(t)$
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
time t (1/Δ)
analytic $\sin^2 t$
Trotter-2 (SV)
shots, N = 1000
(b) spin population is conserved
$\langle Z_1 \rangle$
0.1
0.0
−0.1
0
2
4
6
8
10
time t (1/Δ)
$\pm 2/\sqrt{N}$
SV: $|\langle Z_1 \rangle| < 10^{-15}$
shots, g = 0
shots, g = 2